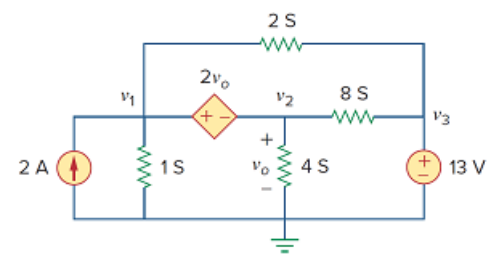


Problem

Determine voltages v_1 through v_3 in the circuit of Fig. 3.65 using nodal analysis.

Figure 3.65:



Step-by-step solution

Step 1 of 4

The given circuit is,

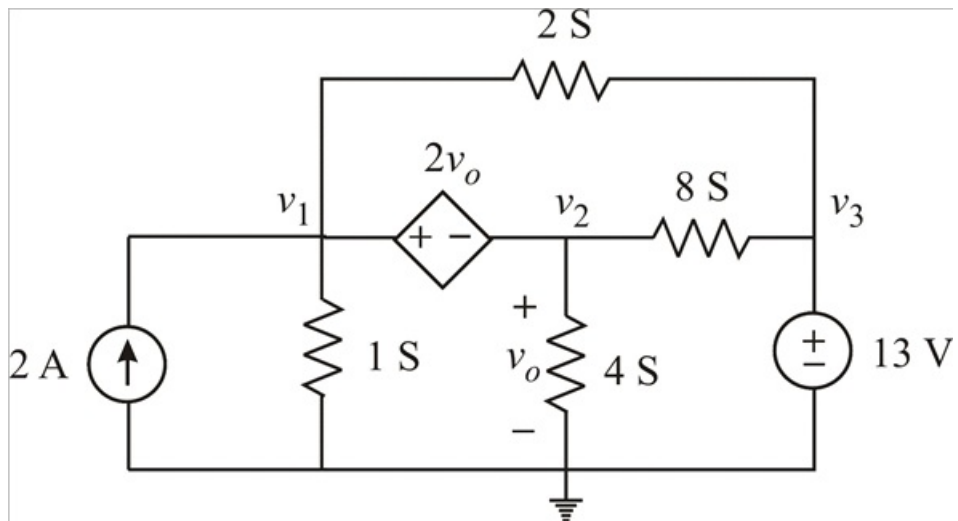


Figure 1

Step 2 of 4

Nodes 1 and 2 form a supernode,

$$v_1 - v_2 = 2v_o$$

Since $v_2 = v_o$.

$$v_1 - v_o = 2v_o$$

$$v_1 = 3v_o \quad \dots\dots\dots(1)$$

Apply KCL at the super node.

$$-2 + v_1(1) + (v_1 - v_3)(2) + (v_2 - v_3)(8) + v_2(4) = 0$$

Since $v_2 = v_o$.

Substitute v_o for v_2 .

$$-2 + v_1(1) + (v_1 - v_3)(2) + (v_o - v_3)(8) + v_o(4) = 0$$

$$-2 + v_1(1) + (v_1 - v_3)(2) + (v_o - v_3)(8) + v_o(4) = 0$$

$$-2 + v_1 + 2v_1 - 2v_3 + 8v_o - 8v_3 + 4v_o = 0$$

$$3v_1 + 12v_o - 10v_3 = 2$$

From the circuit $v_3 = 13\text{V}$.

Substitute 13V for v_3 .

$$3v_1 + 12v_o - 10v_3 = 2$$

$$3v_1 + 12v_o - 10(13) = 2$$

$$3v_1 + 12v_o = 2 + 130$$

$$3v_1 + 12v_o = 132 \quad \dots\dots\dots(2)$$

Step 3 of 4

Substitute $3v_o$ for v_1 from equation (1) in equation (2).

$$3v_1 + 12v_o = 132$$

$$3(3v_o) + 12v_o = 132$$

$$21v_o = 132$$

$$v_o = \frac{132}{21}$$

$$v_o = 6.286\text{V}$$

From the given circuit,

$$v_2 = v_o$$

Hence,

$$\begin{aligned} v_2 &= v_o \\ &= 6.286\text{V} \end{aligned}$$

Step 4 of 4

Substitute 6.286 V for v_o in equation (1).

$$v_1 = 3v_o$$

$$v_1 = 3(6.286)$$

$$v_1 = 18.858\text{V}$$

Therefore, the voltages are, $v_1 = 18.858\text{V}$, $v_2 = 6.286\text{V}$ and $v_3 = 13\text{V}$.